CIS 3300 · INHERITANCE AND POLYMORPHISM

Unit Map

Which notes, slides, and code examples go together

This unit has four parts. Each part has a set of notes, a short slide deck, and a code example, and they are meant to be used together: the slides introduce the vocabulary, the notes explain each idea in depth, and the code example is where you build and run it. Work through the code tutorial, *Inheritance and Polymorphism: Code Tutorial*, alongside the examples.

1. The Four Parts

Table 1: Notes, slides, and code examples for each part of the unit.

Part	Notes file	Slides	Code example
Part 1: Inheritance Basics	`04_inheritance_basics`	Deck 1	Example 1
Part 2: Polymorphism and Casting	`05_polymorphism_and_casting`	Deck 2	Example 2
Part 3: Abstract Classes, final, and Constructors	`06_abstract_classes_and_constructors`	Deck 3	Example 3
Parts 1 to 3 together	The three sets of notes above	Deck 4	Example 4, the rental agency
Part 4: Multiple Inheritance	`07_multiple_inheritance`	Deck 5	Example 5

Each slide deck's file name starts with its number, for example `Slides_1_Inheritance_Basics`, and Deck 4 is the problem we solve together.

2. What Is in Each Code Example

Each example builds on the one before it, so each folder holds a complete program that runs on its own. If you miss a class, you can pick up from the folder for the example before it.

Table 2: The code example folders and the files in each.

Folder	Files	What it adds
`example1`	`Employee`, `Manager`, `Example1Demo`	A first subclass: `extends`, `super`, and overriding
`example2`	`Employee`, `Manager`, `Executive`, `Example2Demo`	A third level, a final `equals` and `hashCode`, arrays and lists of employees, and casting
`example3`	`Employee` (abstract), `SalariedEmployee`, `HourlyEmployee`, `Manager`, `Executive` (final), `Example3Demo`, `ConstructorOrderDemo`	An abstract superclass, a `final` class, and the order in which objects are built
`example4-starter`	`RentalAgency`, `fleet.csv`	The starting point for the rental problem we solve together
`example5`	The Example 3 classes, plus `Payable`, `Contractor`, `Mentor`, `Trainer`, `Example5Demo`	An interface shared with a class that is not an employee, and two conflicting default methods

3. Addenda

Two short addenda came from questions students asked in class. Both go with Part 1.

Table 3: Addenda to the notes.

Addendum	What it covers
Nested Classes and Subclasses	Where a class is declared compared with what type it is
Could a Static Nested Class Replace a Subclass?	Why access to a class is not the same as being a kind of that class

4. Short Examples That Appear Only in the Notes

A few short examples are printed in the notes but do not have a folder of their own. To run one, copy it into a new Eclipse project.

- **Part 1:** the two-package example of access levels (`Person`, `Staff`, `Report`), which shows `protected` and package-private access.
- **Part 3:** serialization with a superclass that is not serializable (`Person`, `Worker`).
- **Part 4:** the rules for conflicting default methods (`Greeter`, `FriendlyGreeter`, `Person`, `Student`, `Tutor`).